Optimal Structured Product Basket Construction: Multi-Asset Monte Carlo Simulation, Downside CVaR Minimization, and Pareto Frontier Allocation

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Abstract

Private banking, wealth management, and institutional investors frequently utilize structured products to enhance portfolio yields and manage asymmetric market exposures. However, sophisticated allocators diversify across multiple product typologies and issuing institutions to mitigate single-barrier drawdowns and issuer concentration risk. We present a rigorous quantitative framework designed to construct and optimize bespoke *baskets of structured products*. By integrating correlated multi-asset Geometric Brownian Motion (GBM) diffusion, path-dependent Monte Carlo payoff resolution, Dirichlet portfolio sampling, and a mathematically consistent Conditional Value-at-Risk (CVaR_α) optimization routine, our framework isolates Pareto-optimal allocations that maximize downside protection for any specified target return. This optimization engine provides an institutional-grade advisory differentiator, transforming individual structured product issuance into bespoke, risk-minimized investment portfolios.

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1 Context & Commercial Rationale

1.1 Client Allocation Dynamics & Issuer Diversification

Within private wealth and institutional asset management, structured products serve as a cornerstone asset class, providing conditional capital protection, enhanced yield generation, and tailored equity index exposures. In practice, prudent capital allocation precludes concentrating principal within a single standalone instrument due to three primary risk factors:

1. **Underlying & Timing Risk:** Isolated autocallable or participation products concentrate barrier vulnerability into a singular maturity profile, observation calendar, and strike level.
2. **Counterparty Credit Risk:** Fiduciary mandates necessitate distributing capital across multiple Tier-1 investment banks to mitigate counterparty default exposure.
3. **Payoff Complementarity:** Strategically blending defensive capital-guaranteed instruments, yield-generating autocallables, and delta-one trackers produces a more resilient cash-flow profile across diverse macroeconomic regimes.

1.2 The Value Proposition: Algorithmic Basket Optimization

Historically, portfolio advisors recommended structured product combinations relying on heuristic approximations. This framework systematically replaces manual discretion with a mathematically robust optimization engine.

By modeling the joint correlation structure of underlying equity benchmarks and generating highly dimensioned macroeconomic pathways, the platform isolates the allocation weights that minimize downside exposure (Probability of Capital Loss, Tail CVaR) for any target Internal Rate of Return (IRR). This optimization unlocks a definitive commercial advantage:

- **Optimized Ticket Sizing:** Incentivizes holistic multi-product tranche subscriptions over fragmented, single-ticket trades.
- **Mathematical Rigor:** Equips allocators with transparent, institutional-grade risk analytics (VaR, CVaR, scenario-weighted cash-flow trajectories).
- **Bespoke Risk Profiling:** Seamlessly integrates customized constraints regarding duration, underlying exposure limits, and capital preservation thresholds.

2 Mathematical Modeling & Multi-Asset Diffusion

2.1 Synthetic Decrement Indices & Issuer Hedging

In structured product manufacturing, issuing investment banks must hedge their structural dividend exposure over long maturity horizons. Because future dividend distributions are uncertain, pricing desks typically reference decrement indices rather than standard Price Return (PR) or Total Return (TR) benchmarks.

A decrement index synthetically subtracts a fixed synthetic dividend linearly on a daily basis from the underlying performance. For a fixed-point decrement, the mathematical index adjustment at daily step τ follows:

$$S_{\text{dec}}(\tau) = S_{\text{dec}}(\tau-1) \cdot \frac{S_{\text{TR}}(\tau)}{S_{\text{TR}}(\tau-1)} - D \cdot \frac{\Delta\tau}{360} \tag{1}$$

where S_{TR} represents the underlying Total Return index, D is the annualized decrement value, and $\Delta\tau$ is the day-count fraction. Our optimization engine fully integrates this dynamic by ingesting or synthetically back-calculating decrement tracks prior to initiating the multi-asset GBM simulation.

2.2 Historical Parameter Estimation

Let K_{assets} denote the number of underlying market indices. For each asset $i \in \{1, \ldots, K_{\text{assets}}\}$, we analyze daily price time-series $\{S_{i,\tau}\}_{\tau=0}^{K}$.

The daily logarithmic returns are defined as:

$$r_{i,\tau} = \ln\left(\frac{S_{i,\tau}}{S_{i,\tau-1}}\right), \quad \tau = 1, \ldots, K \tag{2}$$

Assuming A trading days per annum (typically $A = 252$), the annualized drift μ_i and annualized volatility σ_i are calibrated over a rolling historical observation window:

$$\bar{r}_i = \frac{1}{K}\sum_{\tau=1}^{K} r_{i,\tau} \tag{3}$$

$$\mu_i = A \cdot \bar{r}_i + \frac{1}{2}\sigma_i^2 \tag{4}$$

$$\sigma_i = \sqrt{A \cdot \frac{1}{K-1}\sum_{\tau=1}^{K}(r_{i,\tau} - \bar{r}_i)^2} \tag{5}$$

The empirical cross-asset correlation matrix $R \in \mathbb{R}^{K_{\text{assets}} \times K_{\text{assets}}}$ is derived from pairwise sample covariances. To ensure numerical stability and strict positive semi-definiteness, R is regularized via spectral eigenvalue clipping with subsequent unit-diagonal renormalization.

2.3 Cholesky Decomposition & Continuous GBM Dynamics

Given that the empirical correlation matrix R is symmetric positive-definite, it yields a unique lower-triangular Cholesky factor L_{chol} such that $R = L_{\text{chol}} L_{\text{chol}}^T$.

Let $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, I)$ represent a vector of independent standard normal variables. The vector of correlated standard Gaussian innovations is computed via $\mathbf{Z}_t = L_{\text{chol}}\boldsymbol{\epsilon}_t$. Under the physical probability measure $\mathbb{P}$, each underlying asset follows a correlated Geometric Brownian Motion:

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dW_i(t), \quad \text{with } d\langle W_i, W_j\rangle_t = R_{i,j} dt \tag{6}$$

Discretizing the time horizon $[0, T_{\text{horizon}}]$ into uniform increments of size $\Delta t = 1/A$, the exact discrete solution for scenario n and step t is formalized as:

$$S_{i,t}^{(n)} = S_{i,t-1}^{(n)} \exp\left(\left(\mu_i - \frac{1}{2}\sigma_i^2\right)\Delta t + \sigma_i\sqrt{\Delta t} Z_{i,t}^{(n)}\right) \tag{7}$$

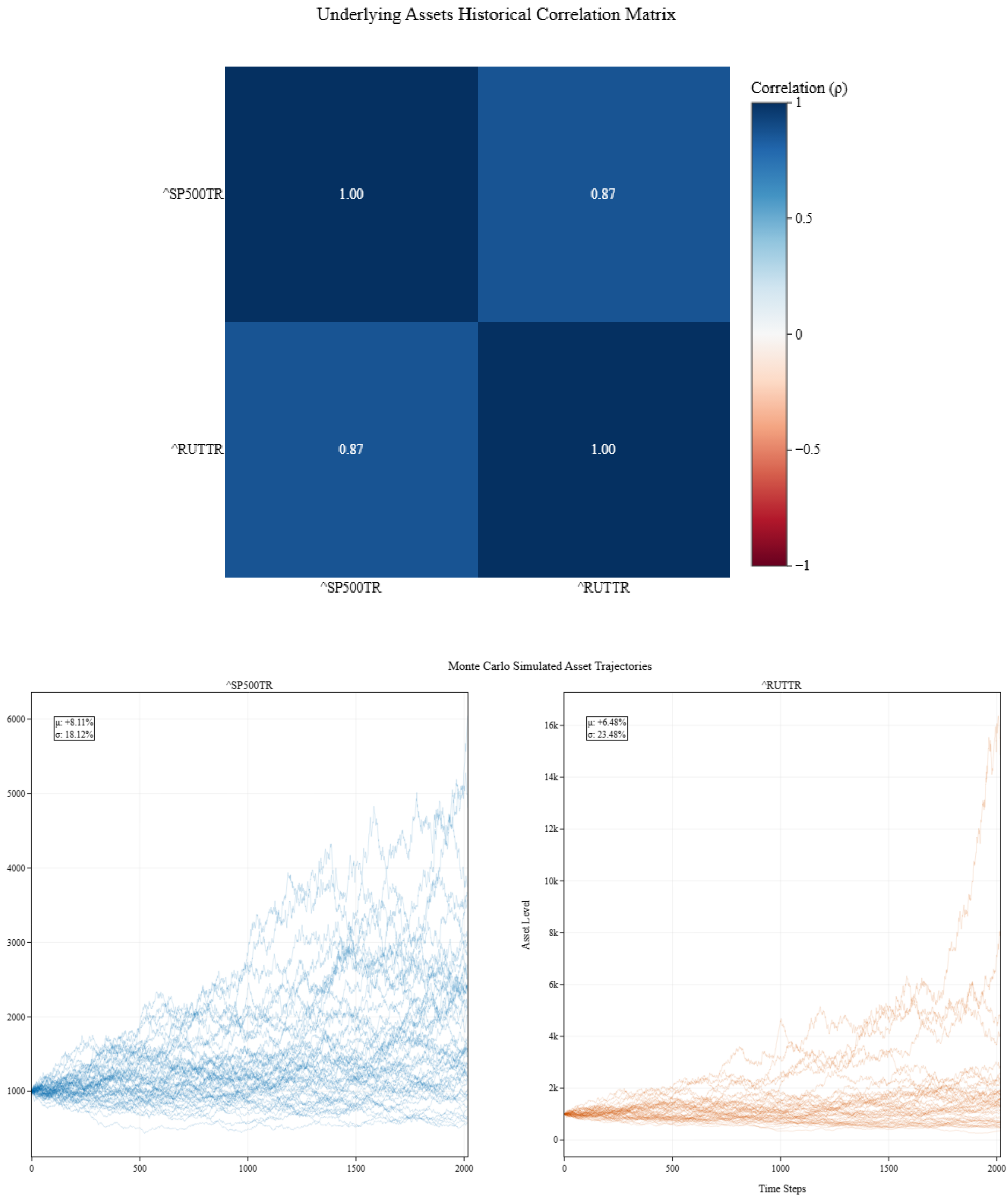

Figure 1: Top: Empirical correlation matrix highlighting strong positive correlation ($\rho = 0.87$) between SP500TR and RUTTR. Bottom: Monte Carlo diffusion fan chart across simulated scenarios over the investment horizon.

3 Structured Product Payoff Architectures

We parameterize a universe of 4 structured product contracts. Contractual cash flows are systematically evaluated path-by-path across all scenarios.

3.1 Preset 1 (P1): Step-Down Autocall

- **Underlying Asset:** Russell 2000 (^RUTTR).
- **Maturity:** 8.0 years, quarterly observation dates.
- **Autocall Barrier:** Initial strike at 100%, stepping down by 1.0% per quarter post-year 1.0.
- **Coupon:** Unconditional 2.0% per quarter (8.0% p.a.) with Memory.
- **Capital Protection:** European barrier at 50% strictly evaluated at terminal maturity.

3.2 Preset 2 (P2): Phoenix Autocall

- **Underlying Asset:** S&P 500 (^SP500TR).
- **Maturity:** 5.0 years, semi-annual observations.
- **Autocall Barrier:** Static 100% strike barrier post-year 1.0.
- **Coupon:** Conditional 4.0% semi-annually (8.0% p.a.) subject to a 75% Coupon Barrier with Memory.
- **Capital Protection:** European barrier at 60%.

3.3 Preset 3 (P3): Capital Guaranteed Asian

- **Underlying Asset:** S&P 500 (^SP500TR).
- **Maturity:** 5.0 years, monthly observations.
- **Payoff Mechanism:** 100% capital guarantee at maturity (0% floor) plus a participation rate of 70% on the average observed index level relative to the initial strike.

3.4 Preset 4 (P4): Delta-One Tracker

- **Underlying Asset:** Russell 2000 (^RUTTR).
- **Maturity:** 3.0 years.
- **Payoff Mechanism:** Linear index performance tracking integrated with an unconditional fixed coupon of 3.5% p.a. distributed at maturity.

3.5 Financial Rationale for Index Allocation

To price these products close to par, the underlying asset's volatility must structurally align with the embedded options logic:

- **Russell 2000 (^RUTTR):** Utilized for the *Step-down Autocall* and the *Delta 1 Tracker*. The higher historical volatility of small caps inflates the premium of the put option sold by the investor, effectively financing the aggressive guaranteed coupons and the highly favorable step-down early redemption probability.

- **S&P 500 (^SP500TR):** Utilized for the *Phoenix Autocall* and the *Capital Guaranteed* product. For the Phoenix structure, lower volatility maximizes the probability of the index remaining above the coupon barrier, ensuring stable yield generation. For the Capital Guaranteed note, lower volatility cheapens the embedded call option, permitting a robust upside participation rate while the zero-coupon bond secures the capital.

4 Optimization Logic: Tensor Algebra & Exact CVaR Formulation

4.1 Primary Tensor Construction & Portfolio Cash Flows

The structural logic executes through centralized vectorized tensor operations to preserve exact scenario variance, correcting the statistical flaw of prematurely averaging underlying portfolio cash flows prior to computing tail risks.

1. **Product Cash-Flow Tensor:** Evaluating every contractual payoff path-by-path yields $\mathbf{C} \in \mathbb{R}^{N\times T\times M}$ where $\mathbf{C}_{n,0,m} = -1.0$ and $\mathbf{C}_{n,t,m}$ is the net cash-flow at step t.
2. **Portfolio Weight Matrix:** We sample L weight combinations $\mathbf{w}_l$ forming $\mathbf{W} \in \mathbb{R}^{L\times M}$ constrained to the unit simplex.
3. **Scenario-Specific Cash Flows:** Applying weight vectors to the product tensor determines the unified net cash flow: $\mathbf{P}_{n,t,l} = \sum_{m=1}^{M} \mathbf{C}_{n,t,m} \cdot w_{l,m}$.

4.2 Scenario Risk Extraction & Re-Averaging Tail Scenarios

Because downside risk is a function of specific adverse market paths, Value-at-Risk must be measured across the N simulated scenarios for a given portfolio. For each target-aligned combination l, we calculate the precise α-quantile of its realized returns, isolating the subset of worst-case market scenarios $\mathcal{N}_{\text{tail},l}$.

Taking a simple arithmetic mean of tail IRRs is mathematically flawed due to the non-linear compounding nature of rates. To derive the exact Conditional Value-at-Risk (CVaR_α):

1. Isolate cash flows $\mathbf{P}_{n,t,l}$ exclusively for scenarios residing in the tail set.
2. Cross-sectionally average the dollar cash flows across these identified worst-case market paths: $\overline{\mathbf{CF}}_{\text{worst-}\alpha,t,l} = \frac{1}{|\mathcal{N}_{\text{tail},l}|} \sum_{n\in\mathcal{N}_{\text{tail},l}} \mathbf{P}_{n,t,l}$.
3. The true expected shortfall $\text{CVaR}_{\alpha,l}$ is defined as the IRR of this aggregated tail cash-flow trajectory. The optimal allocation maximizes this value for a given target return.

5 Allocation Dynamics & Scenario Analysis

Deploying the optimization engine across the defined product universe reveals a structural transition from capital preservation to yield maximization, strictly dictated by the CVaR objective function.

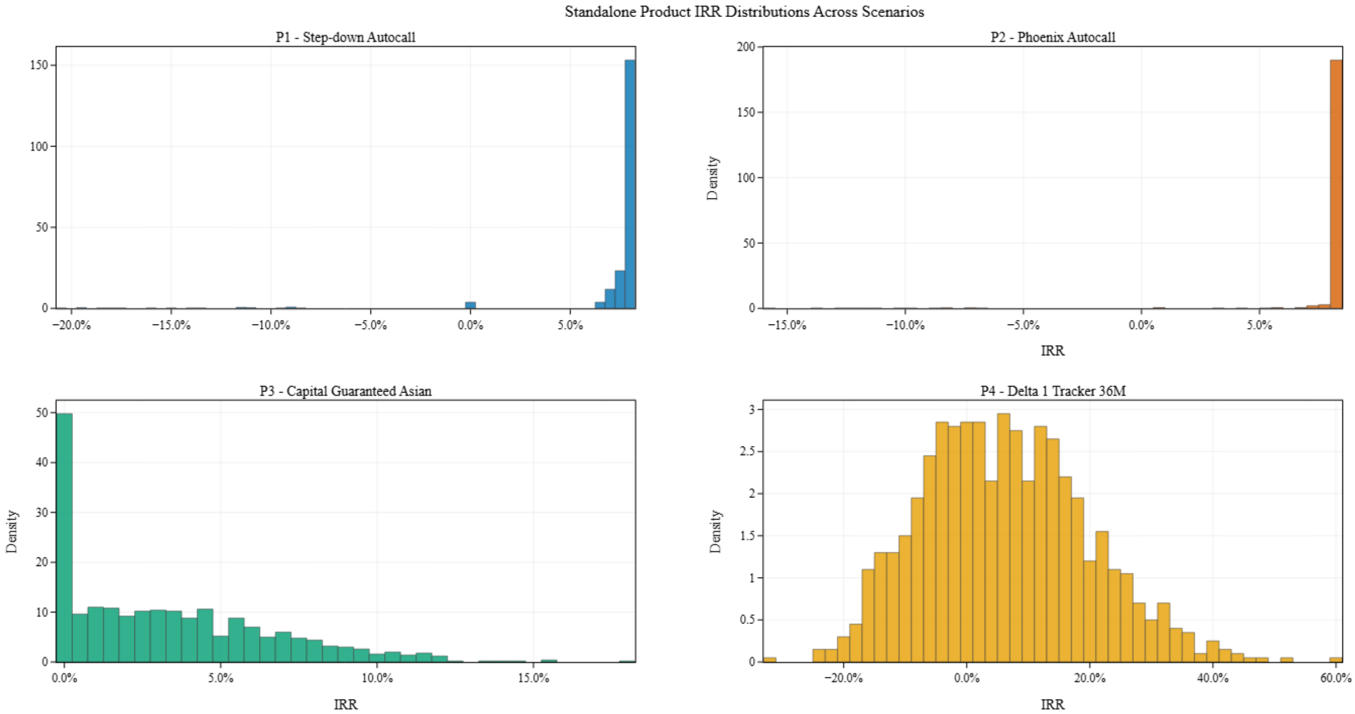

Figure 2: Probability density functions of simulated Internal Rates of Return (IRR) for individual products prior to portfolio optimization.

5.1 Standalone Risk Profiles

The optimizer allocates capital by mapping the asymmetry of product payoffs (Figure 2) against the underlying market trajectories:

- **P3 (Capital Guaranteed):** Exhibits a strict floor at 0% IRR. Indexing to the lower-volatility S&P 500 allows it to capture positive drift without capital risk, rendering it the anchor for low-IRR target portfolios.
- **P2 (Phoenix Autocall):** Functions as the primary yield engine, characterized by a massive density peak at 8.0% (the annualized coupon). However, its long negative tail necessitates careful sizing to prevent severe CVaR degradation.
- **P4 (Delta 1):** Displays a broad, quasi-normal distribution capturing the small-cap risk premium, serving as a linear diversifier without the barrier risks associated with autocalls.
- **P1 (Step-down Autocall):** Despite aggressive coupon generation, its exposure to the highly volatile Russell 2000 leads to frequent capital barrier breaches, resulting in a suboptimal risk/return profile relative to P2. The optimizer consequently marginalizes P1 across all targets.

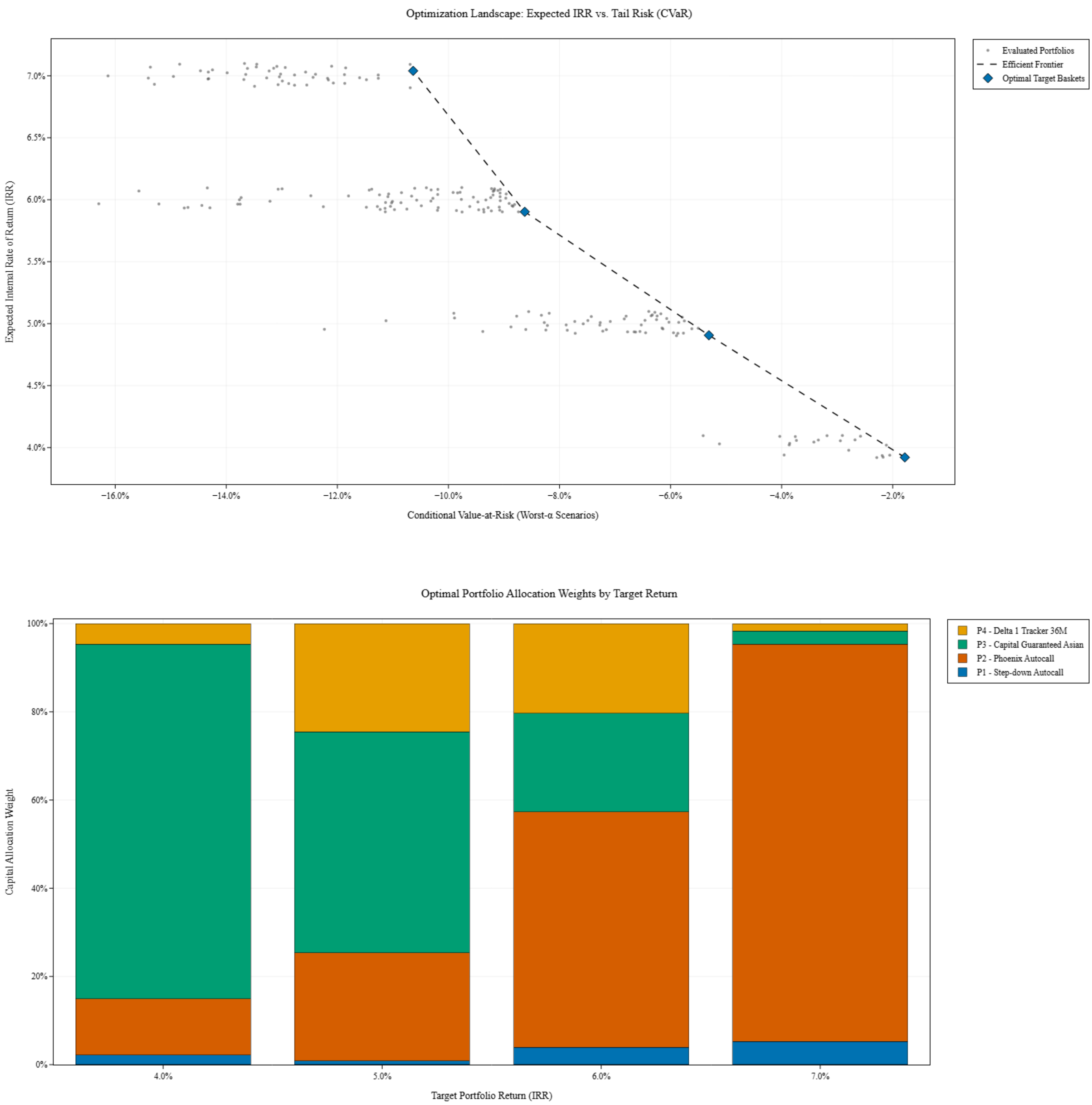

Figure 3: Top: Pareto-efficient frontier maximizing worst-α CVaR. Bottom: Stacked bar chart detailing proportional asset allocations driving the frontier at each target IRR.

5.2 Regime Transition and the Convex Frontier

The weight allocation explicitly shifts through three distinct regimes as return targets escalate:

1. **Defensive Regime (4.0% Target):** The portfolio is heavily dominated by P3 (Capital Guaranteed) at nearly 80% allocation. The optimizer saturates the yield constraint utilizing minimum risk, resulting in an exceptionally robust CVaR near -2.0%.
2. **Transition Regime (5.0% - 6.0% Target):** The protective P3 weight decays linearly, replaced by substantial allocations to P2 (Phoenix) and P4 (Delta 1). To achieve a 6.0%

mean return, the portfolio absorbs directional equity risk and conditional capital barriers, dropping the CVaR to approximately -8.5%.

3. **Aggressive Regime (7.0% Target):** P2 (Phoenix) aggressively cannibalizes the portfolio, representing over 90% of the capital allocation. Reaching this target return natively requires abandoning capital guarantees, effectively transforming the basket into a pure yield engine with a corresponding CVaR penalty (-11.0%).

Crucially, the highly convex shape of the efficient frontier is a direct consequence of the strong positive correlation ($\rho = 0.87$) between the SPX and RUTTR. Absent substantial underlying diversification, the optimizer relies entirely on *payoff diversification* (mixing guaranteed floors with conditional yield), resulting in accelerating downside risk as protective allocations are stripped away.

6 Conclusion & Practical Implementation

The Structured Product Basket Optimizer successfully converges quantitative financial engineering logic with institutional-grade wealth management objectives. By rigorously modeling joint asset correlations and resolving exact tail-scenario cash flows, the engine circumvents the severe limitations of standard Mean-Variance frameworks when applied to asymmetric, non-linear payoffs.

The resulting Pareto frontier provides a mathematically defensible foundation for advisory mandates, proving that optimal structuring requires matching underlying volatility profiles to specific option mechanisms (e.g., financing aggressive coupons via small-cap volatility, while anchoring capital guarantees to large-cap stability). Ultimately, this framework transitions structured product allocation from isolated, heuristic trades into bespoke, dynamically optimized portfolios tailored precisely to client risk-reward parameters.